

Unit 2 - AM

MC - Questions Answered
on D2L

MC	Written	Total
/15	/20	/35

Knowledge/Application

- When dividing $x^4 + 3x^3 - 9x^2 + 6$ by an unknown polynomial, the quotient is $x^3 - 2x^2 + x - 5$ with a remainder of 31. Determine this unknown polynomial.
- For the function $f(x) = 2x^4 + 9x^3 + 9x^2 - x - 3$:
 - Sketch the graph and clearly label the x-and y-intercepts. Show all work!
 - State the interval(s) where $f(x) < 0$. Indicate this on the graph.

Application

- If $f(x) = 4x^3 - 2x^2 + 7x - 12$ and $g(x) = 3x^3 + 5x^2 - 8x - 3$, solve the inequality $f(x) > g(x)$ using the table provided.
- Determine the equation, in standard form, for the family of quartic functions with zeros $x = -\frac{3}{5}$ and $x = 2 \pm 5i$.

Communication

- Is the statement on the right accurate?

$$\frac{5x^3 - 7x^2 - x + 6}{x - 1} = (5x^2 - 2x - 3) + 3$$

If yes, name this type of statement. If not, make the necessary change to the statement for it to be correct and name it.
- Explain how can you determine if $x + b$ is a factor of $x^9 + 3b^2x^7 + 5bx^8 - b^9$? Is it a factor? Justify.

Thinking

- Consider the function $f(x) = x^3 + 4x^2 + kx - 4$. The remainder from $f(x) \div (x + 2)$ is twice the remainder from $f(x) \div (x - 2)$. Determine the value of k.